

International AS and A-level

Chemistry (9620)

CH01 Inorganic 1 and Physical 1

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH01 JUNE 2024

The paper discriminated well: all questions were answered by some students. Students answers to the calculations were, in general, better than the explanations. Some marks were lost because handwriting was poor eg distinguishing between s and g in state symbols. If there is ambiguity, then the mark cannot be awarded. Answers to questions involving practical experiments were often not explained in sufficient detail.

QUESTION 01

Answers to part 1 were very good. Part 2 was not well understood by students: common errors included 10, 8 and 3d. In part 3 nearly three quarters of students gained both marks: common errors included using the A_r for Ge in the periodic table ie 72.6 rather than giving the mass number. Nearly 40% of students gained the mark in part 4: many students did not state that the delocalised electrons move through the structure. Other incorrect answers included mobile ions. Most students scored the mark in part 5; common errors included writing about outer electrons only. The calculation in part 6 was done well. The most difficult mark was finding the mass of one atom in kilograms: some students omitted to divide by 1000, or multiplied by 1000 and a few used the A_r from the Periodic Table instead of the mass number. Most students gained full marks in part 7: the most common error was not giving the answer to 1 d.p. as specified in the question so 65.33 was a common wrong answer.

QUESTION 02

Most students calculated the mean titre correctly in part 1: the most common error was to take the mean of all three titres. About one third of students gained all the marks in part 2: common errors included missing the factor of 4 (conversion of 25 cm^3 to 100 cm^3) and not giving the answer to the required number of significant figures. Students found parts 3 and 4 challenging. Many answers did not describe the practical details needed in part 3 and some answers in part 4 suggested that a conical flask was more accurate (to measure volumes) with no reference to the idea of less chance of losing solution when swirling. In part 5 students often suggested ionic bonding, but then went on to describe covalent bonds or intermolecular forces or described the structure as macromolecular. Nearly 70% of students scored all the marks in the calculation in part 6: often one mark was lost by omitting the 1:2 ratio or by omitting the conversion of celsius into kelvin. Very few students made little or no progress.

QUESTION 03

Most students suggested a correct value for the first ionisation energy: common errors in the explanation included saying that the 2p electron was lower in energy so would need less energy to remove. In part 2 about 47% of students gained both marks; some students lost a mark by referring to the atomic number rather than an increase number of protons or nuclear charge. In part 3 most students drew the diagram correctly: a few students added a lone pair of electrons to the boron. The

mark for the shape was gained by many students but the last mark was gained by fewer students: a common insufficient answer was that the bonds or bond polarities would cancel rather than the dipoles. In part 4 50% of students identified the bond type correctly: common errors in the explanation included the B atom donating a lone pair to fluorine/fluoride ion.

QUESTION 04

Very few students gained both marks in part 1: common errors included heat change or energy change rather than enthalpy change, missing one mole of substance, omitting excess oxygen and not including standard states for all reactants and products. Over half the students scored all the marks in the calculation in part 2: common errors included calculating the moles of propanone incorrectly, using the mass of the reactant rather than the mass of water and omitting the negative sign. Answers to part 3 were good. 43% of students answered part 4 correctly: a common error was that copper is a good insulator. Some insufficient answers were that copper can withstand the high temperatures better or is stronger material than glass. About one third of students gave the correct answer to part 5: the most common incorrect answers were heat loss (given in the question) or the databook value is a mean value. Over half the students gained all the marks in part 6: many students obtained the value + 248 by using the wrong equation and +/- 3832 was a common incorrect answer by using + 1792 instead of - 1792.

QUESTION 05

Most students gained one of the marks in part 1: some students made the melting point of sulfur too high. Students found part 2 challenging: many students lost marks for answers that were not specific enough eg attraction between magnesium and electrons is greater rather than attraction between Mg^{2+} ions and the delocalised electrons is stronger. Some students lost marks by writing about ionisation energy. Very few students answered part 3 well. Incorrect answers included P is macromolecular, P has dipole-dipole forces and covalent bonds are stronger in P molecules. Some students lost marks because they used inaccurate language eg P is larger than chlorine rather than referring to P_4 and Cl_2 or stating they are molecules. Several students said that van der Waals' forces are stronger in phosphorus omitting that these are intermolecular forces. Answers to part 4 were good. Less than 50% of students remembered the equation in part 5: incorrect equations were unbalanced or had the wrong products. In part 6 most students gave the correct sodium halide, but many did not give a correct ionic equation. A few students incorrectly identified the halide as iodine. Answers to part 7 were generally quite good. The reagent was well known: some students gave the name rather than the formula as asked for in the question. Half the students gained one mark in part 8: a common error was giving the acid-base reaction rather than the redox reaction. Common incorrect observations included brown or orange solutions.

QUESTION 06

Most students answered part 1 well: a few students did not balance the equation or had the electrons on the wrong side of the equation. Many students gave a correct equation in part 2: errors included the wrong number of electrons or the wrong number of protons. About 62% of students gave a correct equation in part 3: common errors included unbalanced equations and omitting the H^+ ions. Answers to part 4 were good. In part 5 many students did not state the meaning of reducing agent. The equation was well done: some students lost this mark by adding water or not balancing their equation.